

Problem Statement

- Compare average per-transaction spending between male and female customers during Black Friday.
- Also compare Married vs Unmarried and different age groups.

Key Steps

- Performed EDA: structure, missing values, distributions, outliers.
- Computed group statistics (mean, median, std) for Gender, Marital Status, Age.
- Demonstrated CLT by simulating sampling distributions for different sample sizes.
- Calculated confidence intervals for mean spend at 90%, 95%, 99% for several sample sizes.
- Performed independent t-test to check if mean spending differs significantly between genders.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
import random
```

```
!wget --no-check-certificate https://drive.google.com/file/d/12vBQStSbUugHWlj1ShS0QW8Fv-TdFNUj/view?usp=sharing -O walmart_data.csv
```

```
--2025-10-28 13:58:38-- https://docs.google.com/uc?export=download&id=12vBQStSbUugHWlj1ShS0QW8Fv-TdFNUj
Resolving docs.google.com (docs.google.com)... 74.125.132.101, 74.125.132.102, 74.125.132.100, ...
Connecting to docs.google.com (docs.google.com)|74.125.132.101|:443... connected.
HTTP request sent, awaiting response... 303 See Other
Location: https://drive.usercontent.google.com/download?id=12vBQStSbUugHWlj1ShS0QW8Fv-TdFNUj&export=download [following]
--2025-10-28 13:58:38-- https://drive.usercontent.google.com/download?id=12vBQStSbUugHWlj1ShS0QW8Fv-TdFNUj&export=download
Resolving drive.usercontent.google.com (drive.usercontent.google.com)... 142.250.152.132, 2607:f8b0:4001:c6e::84
Connecting to drive.usercontent.google.com (drive.usercontent.google.com)|142.250.152.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 23027994 (22M) [application/octet-stream]
Saving to: 'walmart_data.csv'
```

```
walmart_data.csv 100%[=====>] 21.96M --.-KB/s in 0.09s
```

```
2025-10-28 13:58:41 (231 MB/s) - 'walmart_data.csv' saved [23027994/23027994]
```

```
data = pd.read_csv('walmart_data.csv')
data.head()
```

	User_ID	Product_ID	Gender	Age	Occupation	City_Category	Stay_In_Current_City_Years	Marital_Status	Product_Category	Purchase
0	1000001	P00069042	F	0-17	10	A	2	0		3
1	1000001	P00248942	F	0-17	10	A	2	0		1
2	1000001	P00087842	F	0-17	10	A	2	0		12

```
data.shape
```

```
(550068, 10)
```

```
data.describe()
```

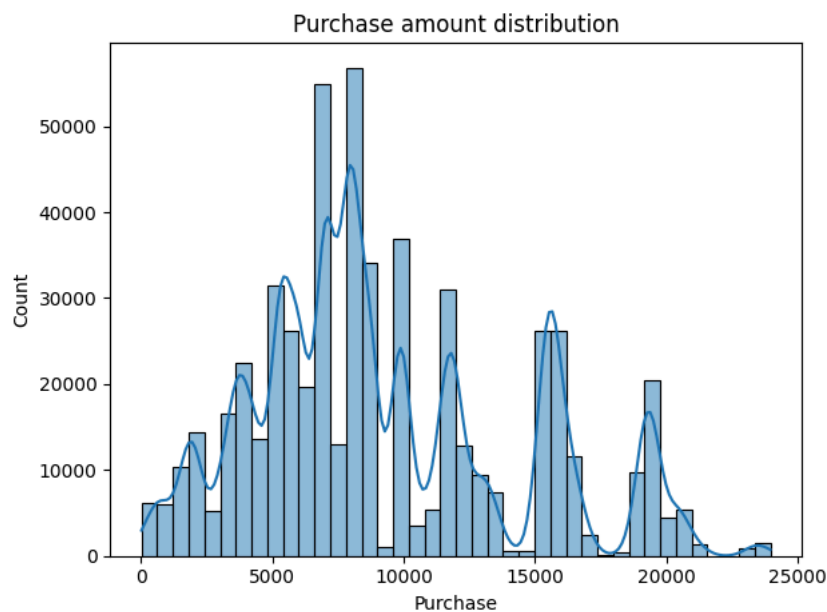
	User_ID	Occupation	Marital_Status	Product_Category	Purchase
count	5.500680e+05	550068.000000	550068.000000	550068.000000	550068.000000
mean	1.003029e+06	8.076707	0.409653	5.404270	9263.968713
std	1.727592e+03	6.522660	0.491770	3.936211	5023.065394
min	1.000001e+06	0.000000	0.000000	1.000000	12.000000
25%	1.001516e+06	2.000000	0.000000	1.000000	5823.000000
50%	1.003077e+06	7.000000	0.000000	5.000000	8047.000000
75%	1.004478e+06	14.000000	1.000000	8.000000	12054.000000
max	1.006040e+06	20.000000	1.000000	20.000000	23961.000000

```
data.isnull().sum()
```

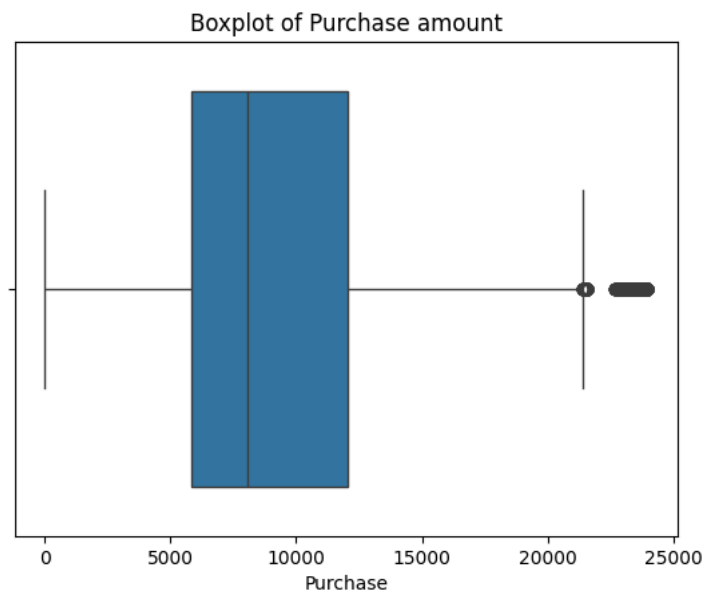
	0
User_ID	0
Product_ID	0
Gender	0
Age	0
Occupation	0
City_Category	0
Stay_In_Current_City_Years	0
Marital_Status	0
Product_Category	0
Purchase	0

dtype: int64

```
plt.figure()
sns.histplot(data['Purchase'], bins=40, kde=True)
plt.title('Purchase amount distribution')
plt.xlabel('Purchase')
plt.ylabel('Count')
plt.tight_layout()
plt.savefig('purchase_hist.png')
plt.show()
```



```
plt.figure()
sns.boxplot(x=data['Purchase'])
plt.title('Boxplot of Purchase amount')
plt.savefig('purchase_boxplot.png')
plt.show()
```

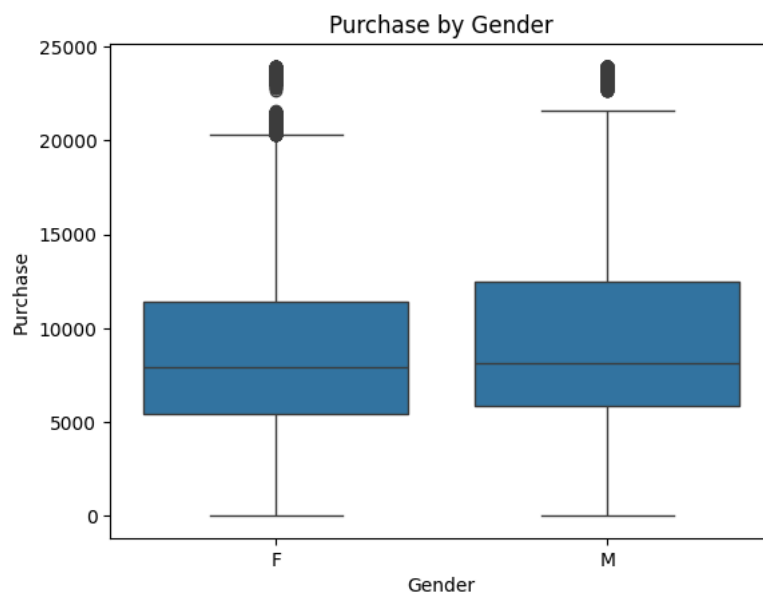


```
Q1 = data['Purchase'].quantile(0.25)
Q3 = data['Purchase'].quantile(0.75)
IQR = Q3 - Q1
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
print(f'Q1={Q1}, Q3={Q3}, IQR={IQR}, lower={lower}, upper={upper}')
```

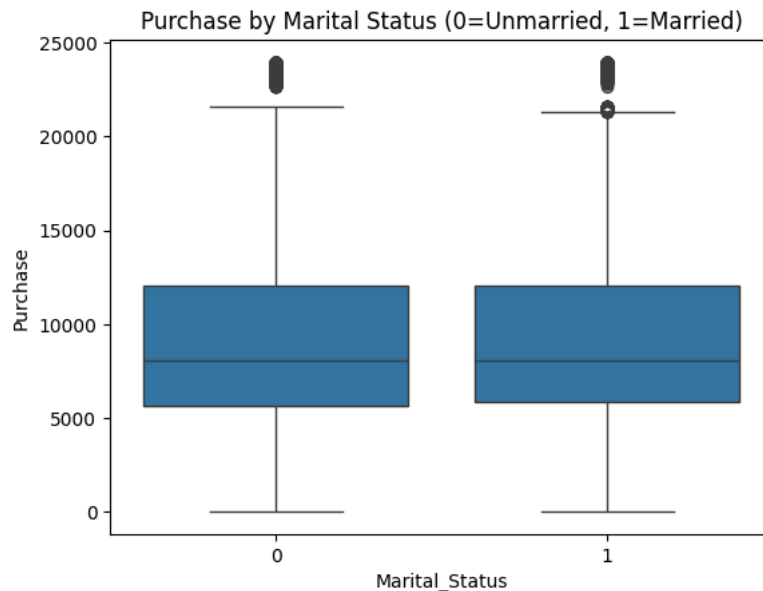
```
outliers = data[(data['Purchase'] < lower) | (data['Purchase'] > upper)]
print('Number of outlier rows by IQR rule:', len(outliers))
```

Q1=5823.0, Q3=12054.0, IQR=6231.0, lower=-3523.5, upper=21400.5
Number of outlier rows by IQR rule: 2677

```
plt.figure()
sns.boxplot(x='Gender', y='Purchase', data=data)
plt.title('Purchase by Gender')
plt.savefig('purchase_by_gender_boxplot.png')
plt.show()
```



```
plt.figure()
sns.boxplot(x='Marital_Status', y='Purchase', data=data)
plt.title('Purchase by Marital Status (0=Unmarried, 1=Married)')
plt.savefig('purchase_by_marital_boxplot.png')
plt.show()
```



```
group_stats = data.groupby('Gender')['Purchase'].agg(['count', 'mean', 'median', 'std']).reset_index()
print(group_stats)

marital_stats = data.groupby('Marital_Status')['Purchase'].agg(['count', 'mean', 'median', 'std']).reset_index()
print('\nMarital status stats:')
print(marital_stats)

age_stats = data.groupby('Age')['Purchase'].agg(['count', 'mean', 'median', 'std']).reset_index()
print('\nAge group stats:')
print(age_stats)
```

Gender	count	mean	median	std
0 F	135809	8734.565765	7914.0	4767.233289
1 M	414259	9437.526040	8098.0	5092.186210

Marital status stats:

Marital_Status	count	mean	median	std
0	324731	9265.907619	8044.0	5027.347859
1	225337	9261.174574	8051.0	5016.897378

Age group stats:

Age	count	mean	median	std
0 0-17	15102	8933.464640	7986.0	5111.114046
1 18-25	99660	9169.663606	8027.0	5034.321997
2 26-35	219587	9252.690633	8030.0	5010.527303
3 36-45	110013	9331.350695	8061.0	5022.923879
4 46-50	45701	9208.625697	8036.0	4967.216367
5 51-55	38501	9534.808031	8130.0	5087.368080
6 55+	21504	9336.280459	8105.5	5011.493996

```
female_purchases = data[data['Gender']=='F']['Purchase'].values
male_purchases = data[data['Gender']=='M']['Purchase'].values
```

```
print('Female sample size available:', len(female_purchases))
print('Male sample size available:', len(male_purchases))
```

```
Female sample size available: 135809
Male sample size available: 414259
```

```
def sample_means(data, n, reps=2000, seed=42):
    np.random.seed(seed)
    means = [np.mean(np.random.choice(data, size=n, replace=True)) for _ in range(reps)]
    return np.array(means)
```

```
sample_sizes = [30, 100, 500, 2000]
sim_results = {}
for n in sample_sizes:
    sim_results[f'F_n{n}'] = sample_means(female_purchases, n, reps=2000)
    sim_results[f'M_n{n}'] = sample_means(male_purchases, n, reps=2000)
```

```
for n in sample_sizes:
    plt.figure()
```

```
sns.histplot(sim_results[f'F_{n}'], kde=True)
plt.title(f'Distribution of sample means (Female) n={n}')
plt.xlabel('Sample mean of Purchase')
plt.savefig(f'clt_female_{n}.png')
plt.show()
```

